John Subba

Systems & Heterogeneous Compute Engineer
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LinkedIn | GitHub

EDUCATION

PURBANCHAL UNIVERSITY Feb/2021 - June/2025
Information Technology Bachelors Nepal

EXPERIENCE

SELF-EMPLOYED | Independent Researcher - GPU Architectures Kathmandu, Nepal | April/2026 – Present

Conducted an intensive deep-dive into GPU architectures, building performance benchmarks and analyzing core algorithms.

Ported foundational CUDA-based vision kernels to the AMD ROCm platform (HIP).

Built a benchmark suite comparing performance, memory bandwidth, and thread occupancy across NVIDIA and AMD hardware.

Gained expert-level understanding of CUDA and ROCm memory hierarchies and warp-level primitives.

RESEARCH MANAGEMENT CELL | AI Project Lead & Technical Mentor Kathmandu, Nepal | May/2026 – June/2026

Led end-to-end architecture for 3 production-grade AI systems.

Architected RAG pipeline with local LLM and real-time 3D avatar rendering for AI Receptionist.

Deployed a local Vision-Language Model (VLM) for real-time feedback on physical tasks (e.g., yoga posture correction) for VLM Posture Corrector.

Built an LLM-based chatbot with real-time web navigation and search capabilities for Autonomous Web Agent.

Provided technical mentorship, conducted architecture reviews, and established code quality standards across all teams.

HIMALOGIC | GPU-Accelerated Computer Vision Engineer (AR/VR Systems) Kathmandu, Nepal | October 2025 – April 2026

Low-level optimization of real-time AR/VR computer vision pipelines, emphasizing GPU acceleration, memory efficiency, and cross-platform execution under strict latency and power constraints.

- Ported OpenCV SIFT pipeline to GPU: implemented Gaussian pyramid, DoG, and scale-space extrema on CUDA; optimized descriptor testing with rapid CUDA kernels.
- Replaced failed KD-tree FLANN matcher with brute-force + voxel-grid approach, achieving target latency and memory footprint on mobile devices.
- Developed Metal backend (iOS, Objective-C++) alongside CUDA backend; implemented native Android Camera2-to-C++ bridges with near-zero-copy buffers.
- Tackled heterogeneous compute challenges across Metal, CUDA, Vulkan, OpenGL ES 3.1/3.2, and WebAssembly; optimized threadgroup sizing and memory placement for mobile power constraints.

Achieved sub-15ms inference on mobile GPU targets with 30% reduction in memory bandwidth.

Tech Stack: C++17/20, OpenCV, Metal Shading Language, CUDA, OpenGL ES 3.1/3.2, WebAssembly, Objective-C++.

INDEPENDENT CONTRACTOR | Sensor Fusion Engineer Remote | July 2025 – September 2025

Designed and implemented real-time multi-sensor fusion systems with emphasis on deterministic execution, numerical stability, and memory efficiency.

- Developed state estimation pipelines using Lie group formulations (SO(3), SE(3)) with Kalman filters (KF, EKF, UKF) and nonlinear estimation.
- Integrated IMU, GNSS, LiDAR, RADAR, and camera measurements into unified state-space models for navigation, tracking, and localization across wheeled, aerial, and pedestrian platforms.
- Prototyped SLAM and graph-based optimization pipelines with loop-closure detection and trajectory estimation.
- Leveraged agentic AI for rapid prototyping, compressing multi-month development into 2 months; implemented on ROS2 and custom low-latency pub/sub frameworks with real-time determinism.

  Achieved <10ms sensor fusion latency on embedded targets.

  Tech Stack: C, C++, CUDA, ROS2, Sensor Fusion, Probabilistic State Estimation, SLAM, SO(3)/SE(3), Carla Simulator.

GUARDWARE AUSTRALIA | Windows Kernel Programmer & Security Engineer | Occasional Python Developer
Remote | February 2025 – July 2025

- Developed and optimized kernel-mode drivers in C/C++ for security enforcement and system integrity.
- Performed system debugging and root cause analysis using WinDbg and static analysis tools.
- Automated testing and deployment workflows using Python; enforced code quality standards.
- Developed browser extensions for security hardening.

GUARDWARE AUSTRALIA | Python & C++ Intern
Remote | Jan/2025 – Feb/2025

- Spearheaded complete analysis and transformation of legacy C# system, redesigning from scratch in Python.
- Achieved 40% latency reduction and improved scalability, maintainability, and resource efficiency.
- Refactored legacy modules and eliminated bottlenecks across the codebase.
- Collaborated with cross-functional teams to ensure seamless transition and enhanced reliability.

LAUNCHPAD AI | Research Fellow
October 2024 – Jan 2025

Pioneered privacy-preserving data matching and integration system using advanced cryptographic frameworks.

- Built two-phase system using privacy-enhancing technologies to securely integrate datasets without exposing PII.
- Designed automated pipelines for data evaluation, candidate matching, and validation optimized for large-scale applications.
- Ensured GDPR, CCPA, and global privacy regulation compliance across healthcare, finance, and e-commerce use cases.

SKILLS

Programming Languages	Proficient at analysing code generated via LLM., English (Fluent), C++, C, Python, SQL
Libraries/Frameworks	Tensorflow, Numpy, Pandas, Matplotlib, Pytorch, scikit-learn, scipy, Cmake, LLM Prompting (Intermediate and above), OpenCV, MediaPipe
Tools / Platforms	Git, GitHub, Linux, Cuda, Docker
Databases	MySQL

PROJECTS / OPEN-SOURCE

FACE-MOOD MUSIC PLAYER | Link
Python, MediaPipe, OpenCV, PyQt6, Pygame, Spotify Web API

Real-time affect-aware desktop pipeline using blendshape-first inference and temporal smoothing to estimate emotions from webcam frames. Routes playback through multiple backends (browser-resolved streams, yt-dlp, local libraries) behind PyQt6 control panel and CLI. Session telemetry and cross-platform audio fallbacks included.

SKYRECON (SATELLITE MAP INTELLIGENCE) | Link Python, PyTorch, OpenCV, PySide6, Google Maps API, CUDA

Built desktop application for satellite building intelligence: fetches and stitches Google Maps imagery by coordinates, runs semantic segmentation (ReFineNet, DlinkNet) to detect structures, applies connected-component analysis with population heuristics. Exports structured results via map-driven, image upload, and batch CSV workflows with CPU/GPU inference paths.

CORTEX ASSISTANT | Link Python, PyTorch, Hugging Face Transformers, Vosk, pyautogui, psutil, Linux/Windows APIs

End-to-end desktop assistant with offline wake word detection, local LLM inference (TinyLlama), and JSON-routed OS automation. Implemented parallel Linux and Windows execution backends for audio, power controls, app launch, media, and system metrics. Hardened small-model behavior with keyword fallbacks and intent normalization.

C++ EXPERTISE:

Expert-level modern C++ for production codebases:

- STL & Algorithms: Complexity-aware design, custom allocators.
- Memory Management: RAII, move semantics, smart pointers for clear ownership.
- Templates & Generics: Type traits, SFINAE, concepts (C++20).
- Concurrency: Mutexes, atomics, lock-free data structures, coroutines.
- Architecture: SOLID principles, design patterns, cross-platform development.
- Tooling: Static analysis (Clang-Tidy, PVS-Studio), sanitizers (ASan, UBSan, TSan), GDB/LLDB debugging.

CERTIFICATIONS

- Goldman Sachs - Software Engineering Job Simulation - Forage
- Data Science Fellow - Fellowship.ai
- Electronic Arts - Software Engineering Job Simulation - Forage